

13. The phase sensitive detector

Sometimes we need to generate a *rectified signal*, i.e. a signal where the current only ever flows in one sense. Typically, we will look at the voltage of the *rectified* signal and we find that any of the +ve-going parts of the waveform are preserved and any -ve going parts of the signal have been reflected about the $V = 0$ axis. Figure 13.1 shows this for a sinusoid (lower trace), along with the rectified version (upper trace); mathematically, this is $|\sin(\omega t)|$.

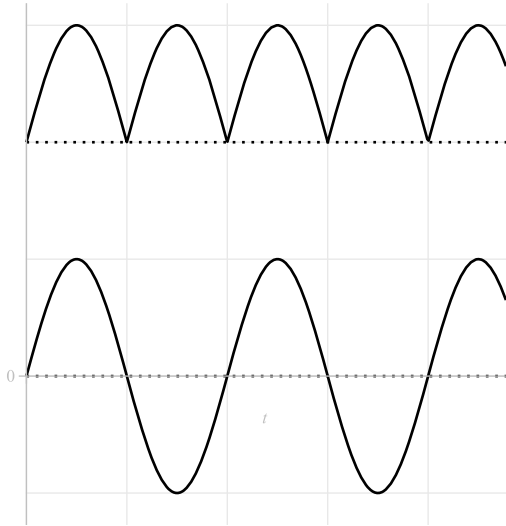


Figure 13.1: Sinusoid (lower trace) and rectified sinusoid (upper trace). In both cases the zero-level is shown by the dotted line.

13.1. Diode rectifier

Since the diode only conducts current in one direction we can make a simple *full-wave rectifier* (which produces an output as per Figure 13.1) using a simple circuit of four diodes, as shown in Figure 13.2.¹⁹

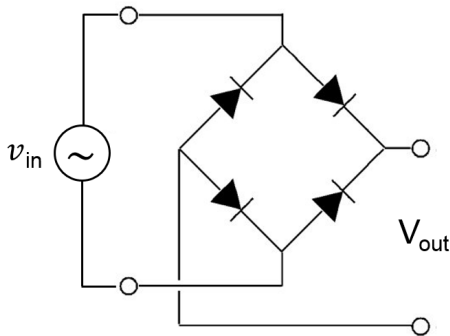


Figure 13.2: Diode bridge full-wave rectifier.

¹⁹See Horowitz & Hill section 1.26.

Example: Power Supply

We might want to build a power supply to charge-up a battery-operated device. This means converting 240 V AC mains to 5 V DC. This is done in three steps. Firstly, a transformer steps the voltage down from 240 V AC to 5 V AC. Secondly, a rectifier gives us a signal per Figure 13.1. Thirdly, the output is *smoothed* using a capacitor. The output is more-or-less a constant voltage, with some *ripple* (see Horowitz & Hill section 1.27); here, we are using the capacitor as a charge-storage device, essentially an integrator.

13.2. Synchronous rectifier

There are problems with diode rectifiers; for example, we get a voltage drop across each diode of about 0.6 V, which means that if we are trying to rectify small AC voltages then we can get a significant reduction in the output amplitude.²⁰ As previously mentioned, op-amps provide a simple, cheap and effective alternative to many traditional techniques, and rectification is no exception. The principle can be seen by looking again at Figure 13.1: the input waveform just needs to be inverted when it swings -ve. That is to say, we need to run it through an amplifier with gain $G = 1$ for +ve voltages and $G = -1$ for -ve voltages (see Figure 13.3). The input signal needs to be multiplied by a square wave at the same frequency (middle trace) to achieve the rectified signal (bottom trace). Op-amps can be used to multiply by a constant gain, so the technique here is to use the square wave to switch the gain between ± 1 . A schematic arrangement for achieving this is given in Figure 13.4. We will not cover the actual circuit in these notes (you can see this in Horowitz & Hill figure 15.37) – it is the principle that is important here: that one can use a reference ancillary waveform at the same frequency as our signal to do some useful processing.

In Figure 13.4 the square wave is called the *reference signal* (we shall see why later) and it is used to control a switch, which flicks the signal from inverted to non-inverted and back again each time the reference signal changes sign. We would describe this as an *edge-triggered switch*, as the transition happens on each rising or falling-edge of the reference signal. This kind of arrangement is quite common, for example a mechanical relay would be fine, though commonly there are electronic switches which will do the job much faster. We will take it as read that the switch is a ready-made component and not be concerned any further with it. The main point to take from this is that *if the reference and input are of the same frequency (and zero phase difference) then the output is a rectified version of the input*.

This requirement that the two inputs be at the same frequency seems quite difficult. However, in many real-life

²⁰Figure 13.1 shows idealised rectification; however, if you look in detail at Horowitz & Hill figure 1.71 you will see that they have taken the small diode voltage drop into account.

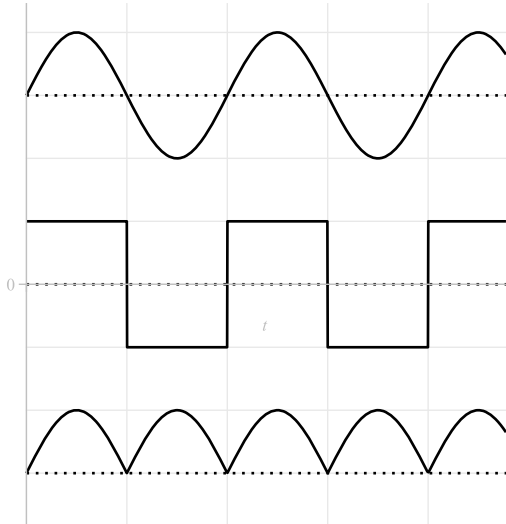


Figure 13.3: Sinusoid multiplied by square wave of same frequency.

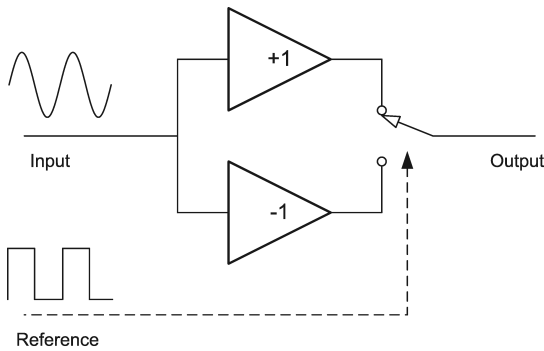


Figure 13.4: Synchronous rectifier.

applications the two signals are derived from a common source so you know they are going to be the same. The fact that this sort of rectifier takes two inputs leads to an application called the *phase detector*, which is a key technique in instrumentation. The phase detector is a synchronous rectifier with a low-pass filter at the output. The importance of the phase detector will be seen when we consider what happens if the two input signals are either out-of-phase, or at slightly different frequencies. First, we need to take a look at the function of the low-pass filter.

13.3. Averaging circuit

We will use a low-pass filter to perform an *average* of the rectified signal – see, e.g. the simple RC circuit in Figure 7.1. We have seen how, for repetitive input signals of high frequency (compared to the cut-off, i.e. $\omega \gg 1/RC$), the low-pass filter *approximates* to an integrator (Table 7.1). If we integrate over one time constant ($\tau = RC$) then

$$V_{out} = \frac{1}{\tau} \int_0^{\tau} V_{in} dt = \langle V_{in} \rangle.$$

This makes sense from a physical point of view.²¹ When the input voltage is greater than the voltage on the capacitor, the capacitor is charging up; when the input is lower then it is discharging. This charge/discharge cycle will be seen as ripple on the capacitor voltage, but for high frequency inputs ($\omega \gg 1/RC$) the ripple is very small and the capacitor voltage will approximate the average input, i.e. $V_{out} \approx \langle V_{in} \rangle$.

13.4. Phase detection

To quote Horowitz and Hill: “*This is a method of considerable subtlety*”, which is their typically understated way of introducing a rather complicated concept. Because of its importance, though, we will go through the topic in some detail. The *phase detector* simply consists of a *synchronous rectifier with a low-pass filter to average the output*. The importance of the phase detector will be seen when we consider what happens if the two input signals are either out-of-phase, or have different frequencies. We will investigate these in turn.

13.5. Behaviour with a phase difference

This is best understood graphically, with the help of Figure 13.5. In all the cases presented here, the two input signals have *exactly the same frequency*. In the first panel, there is zero phase difference and we get a constant positive output which turns out to be $2/\pi$ times the amplitude of the input signal. Inverting the input signal ($\phi = \pi$) gives us a negative output of the same size. For the $\phi = \pi/2$ case we get a zero output and for the “in-between” case of $\phi = \pi/4$ we get an in-between output. Clearly then, the phase detector is highly sensitive to differences in phase between the two signals. For an input signal $V_{in} = A \sin(\omega t + \phi)$ the output of the phase detector is

$$V_{out} = \frac{2A}{\pi} \cos \phi. \quad (13.1)$$

This can be checked by performing the rectification/integration directly as

$$V_{out} = \int_0^{\pi/\omega} V_{in} dt - \int_{\pi/\omega}^{2\pi/\omega} V_{in} dt.$$

In summary then, when the two inputs have the same frequency, the phase detector gives a voltage output which is a function of the phase difference between the two. This in itself is very important and has applications which we will not go into here, because the most useful function of the phase detector becomes apparent when we consider how the circuit behaves as we vary the *relative frequency* of the two inputs.

²¹Consider the Fourier components of the input signal and recall that the filter only lets through the very low frequency components. The rectified signal has a DC component (non-zero value of A_0 , cf. Section 5) which is the long-term average of the signal.

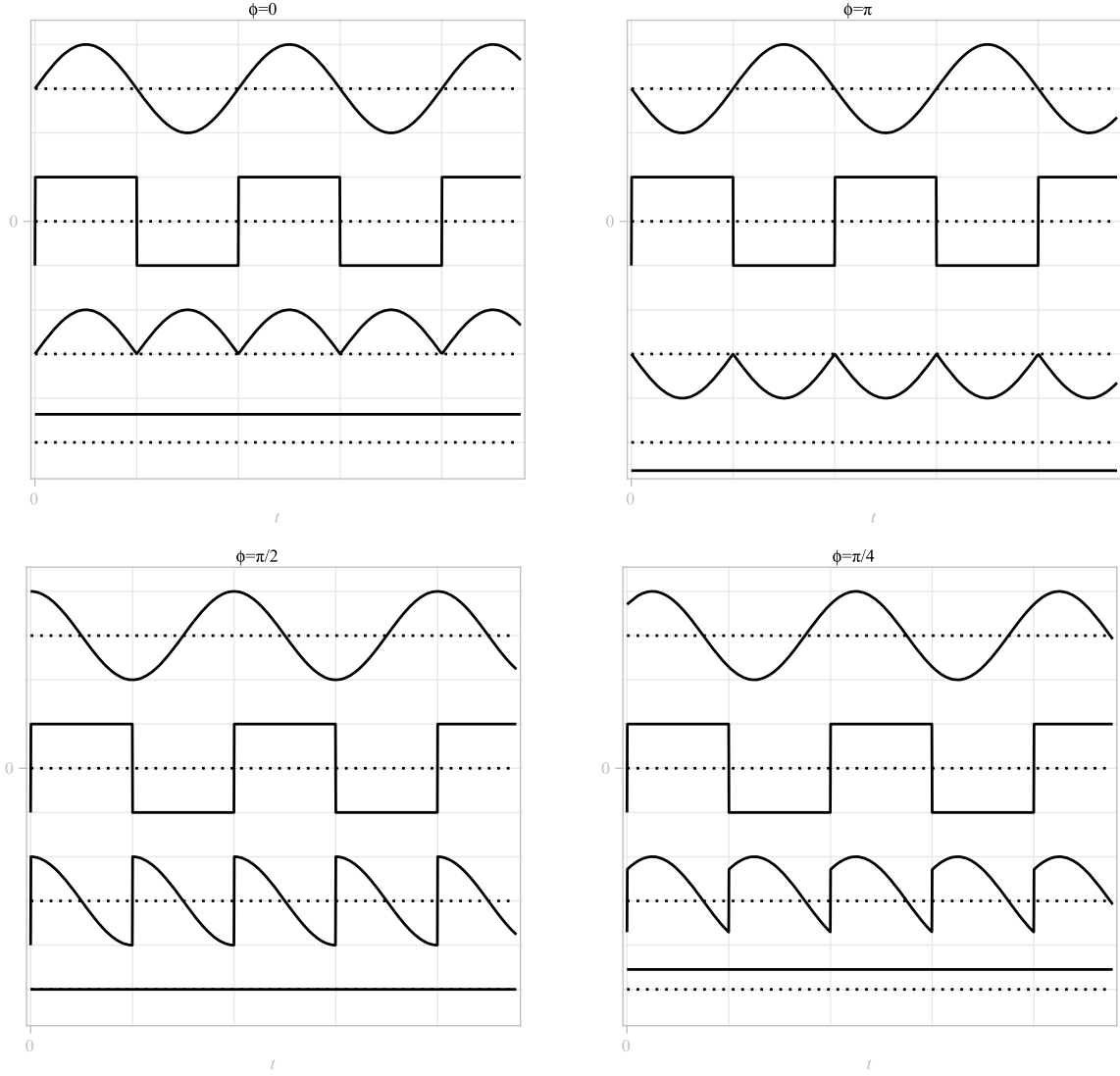


Figure 13.5: Phase detector behaviour with four different values of phase difference ϕ between input and reference signal. Note that in all these figures, the waveforms from top to bottom are: input signal, reference signal, rectified signal and averaged (output) signal.

13.6. Behaviour with a frequency difference

In this situation we define two different frequencies ω_{input} for the input signal and ω_{ref} for the reference. This is illustrated with the four plots in Figure 13.6. For $\omega_{input} = 2\omega_{ref}$ we can see that the output will always be zero. If we choose a ratio such as $\omega_{input} = 1.3\omega_{ref}$ then as long as we average over a sufficiently large number of cycles we always get zero output.²² We need to reduce the ratio to $\omega_{input} = 1.01\omega_{ref}$ before we start to see any kind of output from the detector. It looks like a constant output, but if we plot the signal over a longer time period (bottom-right panel) we can see that it is sinusoidally varying. In fact the output varies as $\cos(t\Delta\omega)$, where $\Delta\omega$ is the difference between the two input signals, *as long as $\Delta\omega$ is a low-enough frequency to pass through the filter.*

We can understand this behaviour by looking at the frequency content of the signals. If the reference frequency is ω and the input frequency is $\omega + \Delta\omega$,

$$\begin{aligned} V_{in} &= A \sin(\omega + \Delta\omega)t \\ V_{ref} &= \frac{4}{\pi} \left[\sin \omega t + \frac{\sin 3\omega t}{3} + \frac{\sin 5\omega t}{5} \right]. \end{aligned}$$

Here we used a Fourier expansion to represent the square wave, and we just need the first three terms of the Fourier series to understand the behaviour. Multiplying the two together, we find the rectified signal (i.e. *before* the filter)

²²For these plots, the averaging period is 10 cycles.

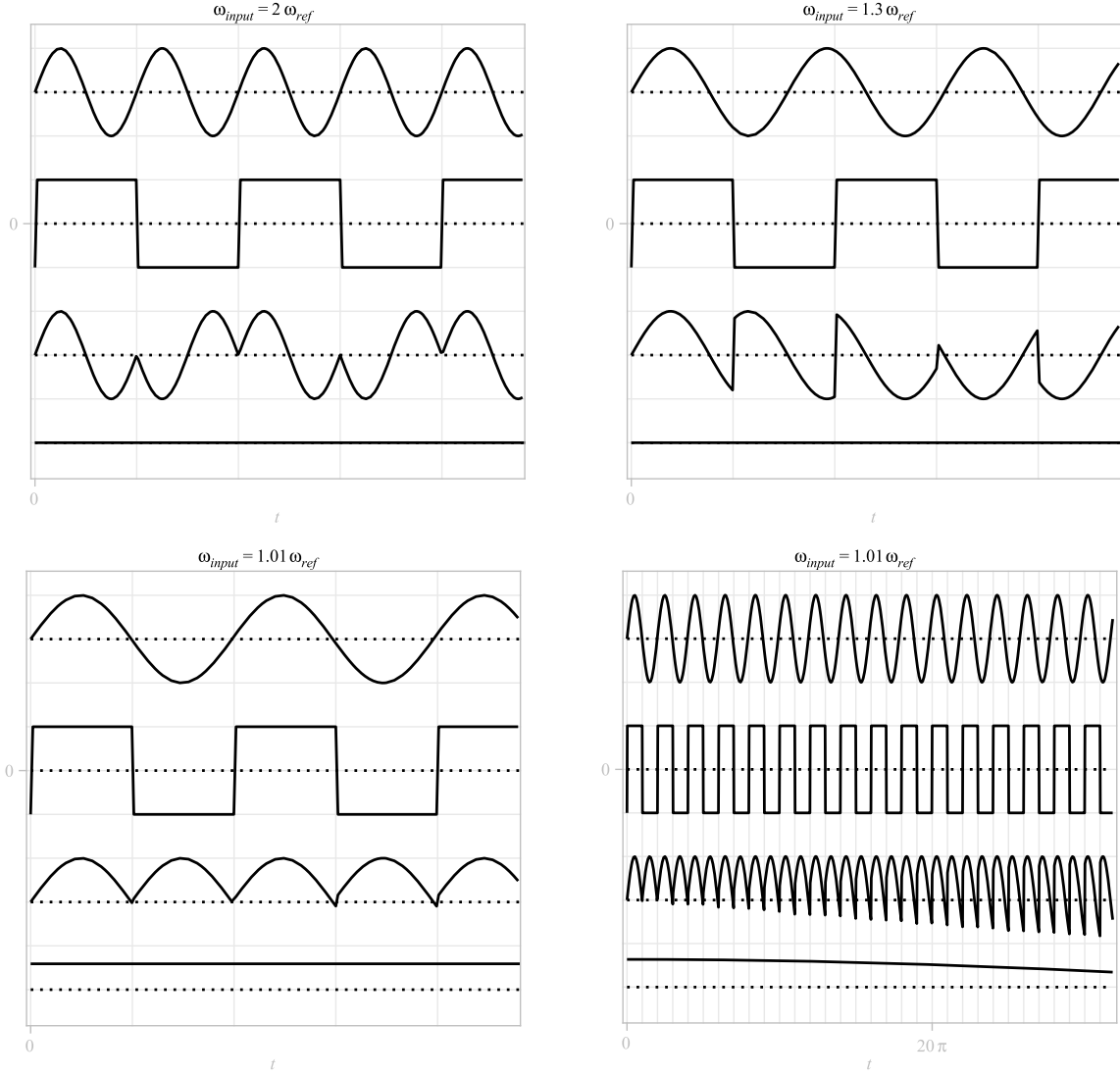


Figure 13.6: Phase detector behaviour where the input signal has a different frequency to the reference frequency. Waveform order in each plot as per Figure 13.5.

is given by²³

$$V_{rect} = \frac{2A}{\pi} [\cos \Delta\omega t - \cos(2\omega + \Delta\omega)t + \frac{1}{3} \cos(\Delta\omega - 2\omega)t - \frac{1}{3} \cos(4\omega + \Delta\omega)t + \frac{1}{5} \cos(\Delta\omega - 4\omega)t - \frac{1}{5} \cos(6\omega + \Delta\omega)t]. \quad (13.2)$$

We have implicitly assumed above that $\omega \gg \omega_c$, and so frequencies ω and above will *not* pass through the filter. But for $\Delta\omega \ll \omega_c$ the first term in equation (13.2) will pass through the filter un-attenuated – i.e. although the filter removes the fundamental frequency of our input signal plus any higher frequency terms, the *difference frequency* relative to the reference signal is slow enough and hence

perfectly able to pass through. Thus, we can define the output as follows:²⁴

$$V_{out} = \begin{cases} \frac{2A}{\pi} \cos(t\Delta\omega) & \Delta\omega \ll \omega_c \\ 0 & \Delta\omega \gg \omega_c \end{cases}. \quad (13.3)$$

We will return to these phase-sensitive techniques in Section 21, the final section in these notes, after a discussion of ‘noise’. The ability to narrow the measurement bandwidth to extremely small values is critical when faint signals of known frequency need to be extracted from a large noise, for example.

²³We use $\sin A \sin B = 1/2[\cos(A-B) - \cos(A+B)]$: the product of two sinusoids is (co-)sinusoids at the difference and sum frequencies.

²⁴Note that we can use a similar argument to demonstrate equation (13.1) too.

14. Linear Systems

We can characterise a physical system by its ability to accept an input parameter and produce an output in response. Our understanding of a sensor, or transducer, is a perfect example. We will now examine this relationship between input and output in more detail.

Systems can be linear or non-linear. A *linear system* is one for which a linear relation exists between cause and effect. Obvious examples include a spring, for which force is directly proportional to elongation, or a resistor, for which voltage and current are proportional through Ohm's law. A counter example might be a diode, which has a very non-linear relationship between voltage and current. However, for many systems such a casual inspection will not tell us whether it is linear or non-linear. To determine whether a system is linear or not we need to examine the mathematical formulation of the input/output relationship. The reason we are interested in this is because, for linear systems, we can use frequency-domain techniques (Fourier and Laplace) to understand and predict the behaviour of really very complex systems.

14.1. Linear time-invariant systems

Any linear time-invariant system, device or component can ultimately be described by a differential equation of the form

$$a_0 x(t) + a_1 \frac{dx(t)}{dt} + a_2 \frac{d^2 x(t)}{dt^2} + \dots + a_n \frac{d^n x(t)}{dt^n} = b_0 y(t), \quad (14.1)$$

where a_n and b_0 are constants, $x(t)$ is the *output* of the system and $y(t)$ is the *input*. Systems which are described by such a differential equation are often termed *Linear Time Invariant* or *LTI* systems. The *time invariance* comes from the fact that the behaviour of the system (its differential equations) does not vary with time, that is to say the behaviour, under a given input, is the same today as it was yesterday, and it will be the same tomorrow.

The input $y(t)$ is also known as the *forcing function* and may be any arbitrary function, for example, it may be a constant (recall the static response), it may be a sinusoid at a given frequency (recall the frequency response), or it may be a transient function such as the unit step or the delta function. LTI systems have two important properties which we explore in turn.

Frequency preservation. If the input to an LTI system is a sinusoid then the output is also a sinusoid of the same frequency, but it may have its amplitude and phase modified (Figure 14.1). Note that this follows directly from equation (14.1). If the input $y(t)$ is a sinusoid, then $x(t)$ must also be a sinusoid at the same frequency – given that all the derivatives of x are themselves sinusoids at the same frequency.

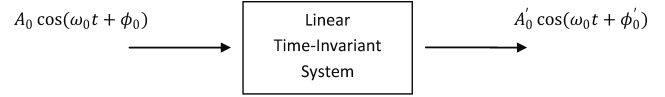


Figure 14.1: Frequency preservation of LTI systems.

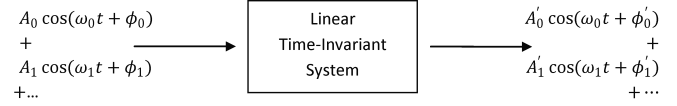


Figure 14.2: Superposition property of LTI systems.

Superposition. If the input to an LTI system is the sum of several sinusoids then the output of the system is the sum of the system's response to each individual sinusoid in isolation (Figure 14.2). Note that, for each frequency component, we can read-off the change in amplitude and phase from the Bode plot. A more sophisticated expression of this concept is given below.

14.2. Use of Fourier techniques

The properties of superposition and frequency preservation allow us to apply useful Fourier techniques to real-world problems involving LTI systems. Any real-world signal can be Fourier-decomposed into a sum of sinusoids of suitable frequency, phase and amplitude, and we can follow the propagation of each sinusoid through an LTI system (each component will have its phase and amplitude modified) and then sum the output sinusoids to recover the response of the system to the original signal. We can see from these properties that *if we add several LTI systems together in series, then the resultant system is also LTI*. We will see how to apply these ideas later on in the course.

The most common systems we come across in instrumentation are quite well described by rather simplified versions of equation (14.1). It is worth briefly revising how simple zero-, first- and second-order systems respond to a simple forcing function such as a step $u(t)$ or an impulse $\delta(t)$. The aim of this section is to briefly run through some of the background mathematics, particularly for the second-order system. Note that the second-order system is the damped harmonic oscillator which you will already have covered a number of times before in other courses. What we typically find is that, whilst the differential equations describing isolated zero, first- and second-order systems are generally quite tractable mathematically, it can be extremely difficult to both formulate and solve a differential equation describing the behaviour of a more complex instrument composed of several distinct elements. However, as suggested above there are solutions to this based on Fourier techniques, which we will come on to later.

14.3. Zero-order systems

Zero-order systems are described by an equation of the form

$$a_0 x = b_0 y,$$

where a_0 and b_0 are constants, $y(t)$ is the input or forcing function applied to the system, and $x(t)$ its output. They thus represent the most trivial of LTI system. This type of system responds immediately to a changing input, and has no possibility for delay, oscillation, or dependency on past behaviour. Zero-order systems do not care what has happened to them in the past, only what the state of the input is now. This type of behaviour would be ideal for a sensor, we would get infinitely fast time response, and no problems with oscillation, overshoot, settling times and so forth. Unfortunately, such systems are extremely rare in real life instruments, but we will look briefly at them to provide us with some scene-setting examples. A good example of a zero-order system is a variable resistor (Figure 14.3). The input to the system $y(t)$ is the position of a

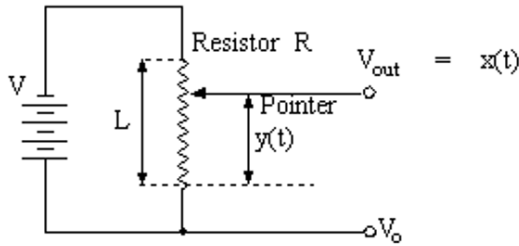


Figure 14.3: Variable resistor (or potential divider) used to measure displacement.

pointer along the resistor, and the output $x(t)$ is a voltage V_{out} that appears at the end of the pointer arm. For the zero-order case V_{out} exactly follows $y(t)$, and at any time,

$$x(t) = \frac{V_0}{L} y(t),$$

where V_0 is the voltage across the full length L of the resistor. Any change in $y(t)$ gives an immediate corresponding change in x . As soon as y stops changing, so does x . This, however, is unphysical for a real-world system, and in our example effects such as the slight flexing of a mechanical pointing arm as it moves or stray inductive or capacitive effects would stop x exactly tracking y . More generally, we will have to formulate and solve a differential equation to find the time response of a system to any given input. However, under certain conditions we can approximate the behaviour of some components as zero order, for example a resistor ($v = iR$) or an idealised amplifier (the output is simply an amplified version of the input).

The zero-order system is characterised by the *gain*, or more generally the *static sensitivity* K_s ; in our example,

$$K_s = \frac{b_0}{a_0} = \frac{V_0}{L} \quad [\text{V/m}].$$

14.4. First-order systems

First-order systems are described by a differential equation of the form

$$a_1 \frac{dx}{dt} + a_0 x = b_0 y. \quad (14.2)$$

The rate of change of the output $x(t)$ is thus proportional to the difference between the system's current state (at a given point in time) and the value of the forcing function. To solve a differential equation of this type we generally try a test solution of the form $x(t) = Ae^{st} + C$, where A , s and C are constants. We then apply boundary conditions to find the values of these constants. An example of this method, as applied to the RC circuit, was given in Section 7.

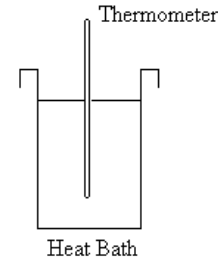


Figure 14.4: Thermometer in heat bath.

By way of example, consider the thermometer in Figure 14.4, which is initially at temperature $y = \theta_0$ and plunged at time $t = 0$ into a heat-bath of temperature θ_R . The rate of heat flow into the thermometer is proportional to the *difference* between the temperature of the thermometer and the heat reservoir θ_R . Our boundary conditions are that at $t = 0$ the thermometer is at its starting temperature θ_0 , and for $t \rightarrow \infty$ it reaches the same temperature as the heat reservoir. The solution is

$$x(t) = K_s \left[\theta_0 + (\theta_R - \theta_0)(1 - e^{-t/\tau}) \right].$$

If $x(t)$ is the length of the mercury column in mm and the θ parameters are temperatures in $^{\circ}\text{C}$, then K_s is the static sensitivity in $\text{mm}/^{\circ}\text{C}$. In terms of the original parameters of equation (14.2), we again find that

$$K_s = \frac{b_0}{a_0}.$$

Note that τ is a *time constant* dependent on the conductivity and thermal mass of the thermometer; τ is the time taken for the system to reach 63.2% of its steady state response, after the application of a step input. This type of time response is very common in instrumentation, see the handout on the AD950 temperature sensor for an example. We often refer to the sort of plot shown in Figure 14.5 as an “RC curve”, as it follows the form of the voltage

across a capacitor as we charge it up from a DC source through a resistor (see Section 7). The system initially responds rapidly to a step input, as the difference between its starting point and the steady state value is large. As time progresses, the difference between steady state and actual value becomes smaller, and the rate of change falls off correspondingly. After a sufficiently long time the system reaches its steady state value, though for a real sensor, the time taken to reach 95% of its steady state value, or just the time constant τ will be given on the datasheet.

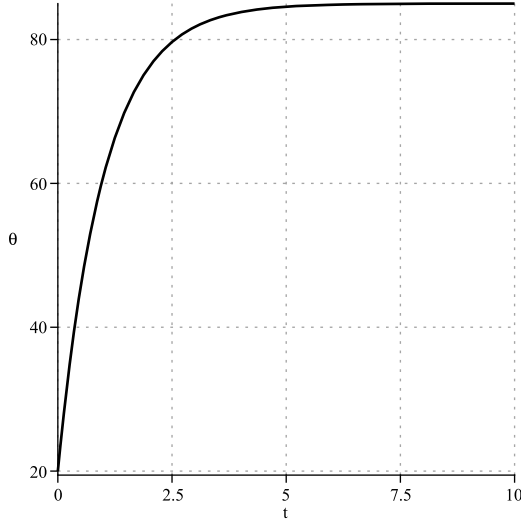


Figure 14.5: Room temperature thermometer plunged into a nice hot cup of tea.

Expressing the time constant τ in terms of the parameters in equation (14.2), we find

$$\tau = \frac{a_1}{a_0}.$$

Referring back to Sections 1 and 2, recall that when we need to calibrate a first-order instrument such as a thermometer we need to hold the input constant and wait a sufficient number of time-constants for the output to settle before taking a reading. As a quick exercise: how many time constants must elapse before the output reaches 99.5% of the steady-state? After this time, we can assume $dx/dt = 0$ and so the output:input ratio gives us K_s .

14.5. Second-order systems

Second-order systems are described by a differential equation of the form

$$a_2 \frac{d^2x}{dt^2} + a_1 \frac{dx}{dt} + a_0x = b_0y. \quad (14.3)$$

This is essentially the defining equation of a damped simple harmonic oscillator. A good example of this type of system in instrumentation is a spring balance (Figure 14.6, although there are many non-mechanical physical analogues). Second-order systems can show the type

of delayed time-response to a step input that we saw in first-order systems, but they can also oscillate after a step $u(t)$ or impulse $\delta(t)$ input. We can begin by summing all the forces that act on the mass m . There is a gravitational force mg , a force due to the extension of the spring kx , a velocity dependent damping term $c dx/dt$ and an acceleration term $m d^2x/dt^2$. Summing all the forces on the mass gives

$$m \frac{d^2x}{dt^2} + c \frac{dx}{dt} + kx = mg. \quad (14.4)$$

There are several limiting cases of the behaviour of this system that are easy to find. For the steady state case there are no changes as a function of time and $d^2x/dt^2 = dx/dt = 0$, the response is the same as a zero-order system. The static response of the system is thus $mg = kx$; therefore, the static sensitivity is

$$K_s = \frac{b_0}{a_0} = \frac{g}{k}.$$

For the spring balance example, we intuitively know that we have to wait for any oscillations to be damped out before we take a reading, and therefore in the steady-state case we find $x = mK_s$ where x is the extension of the spring.²⁵

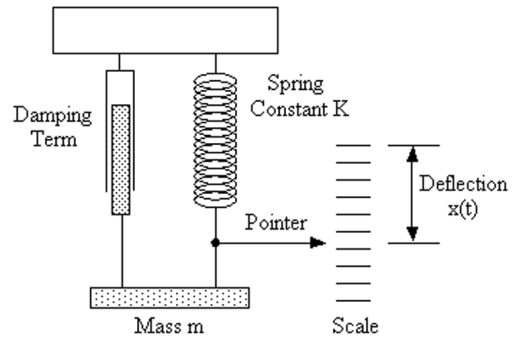


Figure 14.6: Spring balance.

14.5.1. Solution to the second-order differential equation

To understand the behaviour of the second order system it is convenient to re-write equation (14.5) as

$$\frac{d^2x}{dt^2} + 2\xi\omega_0 \frac{dx}{dt} + \omega_0^2x = g, \quad (14.5)$$

where ω_0 is the *natural frequency* of the system and ξ is the *damping ratio*:

$$\begin{aligned} \omega_0^2 &= \frac{a_0}{a_2} = \frac{k}{m} \\ \xi &= \frac{a_1}{2\sqrt{a_0a_2}} = \frac{c}{2\sqrt{mk}}. \end{aligned}$$

²⁵Note that for the typical balance this linear displacement x is translated into an angle on a dial by some arrangement of gears.

The exact form of the solution²⁶ to equation (14.5) depends on the numerical value of the damping ratio; however, in general

- For $\xi = 0$ equation (14.5) becomes that for a simple harmonic oscillator, i.e. the solution is sinusoidal;
- For $\xi < 1$ the solution is an exponentially-decaying sinusoid;
- For $\xi > 1$ the solution is exponential-decay with no oscillation (similar to the first order system).

14.5.2. Behaviour of the damped harmonic oscillator

Returning to our spring balance example, we find that for the zero damping case ($c = 0$, thus $\xi = 0$) any disturbance of the system will result in free running oscillations at a frequency ω_0 . This type of free running behaviour is almost always a bad thing for an instrument. After all, we want to use it to measure the forcing function. We will, however, see that allowing a little oscillation to occur can actually be a good thing for an instrument, improving its time response at the cost of some small and rapidly damped oscillations after an abrupt change to the input.

The behaviour is summarised in Figure 14.7, which gives the response of the balance to a step change input, i.e. suddenly placing the mass m on the balance at time $t = 0$. The plots are normalised to show a unit-response and give 6 different values of $\xi = 0, 0.01, 0.1, 0.5, 1$ and 3 . For $\xi = 0$ we have free running at ω_0 . This would be disastrous in a real balance since we want to use the *static response* to determine the mass, i.e. we want the balance to settle. For $\xi = 0.01$ and 0.1 the system is *underdamped*. At $\xi = 0.6$ is the kind of behaviour we might expect from a real spring balance as it is probably the optimum: some oscillation but rapidly reaching equilibrium. A car's suspension has this sort of behaviour. For $\xi = 1$ the oscillator is *critically damped* – there is no oscillation but the system takes quite a time to reach the equilibrium. The spring closers for doors might be designed for this sort of behaviour. $\xi = 3$ is the *over-damped* case; the behaviour is reminiscent of the first-order system.

²⁶The general method consists of finding the solution for both forced and unforced cases and adding them together (this works because the system is linear). This method of solution is not part of the course; however, you can follow it in Poularikas and Seely.

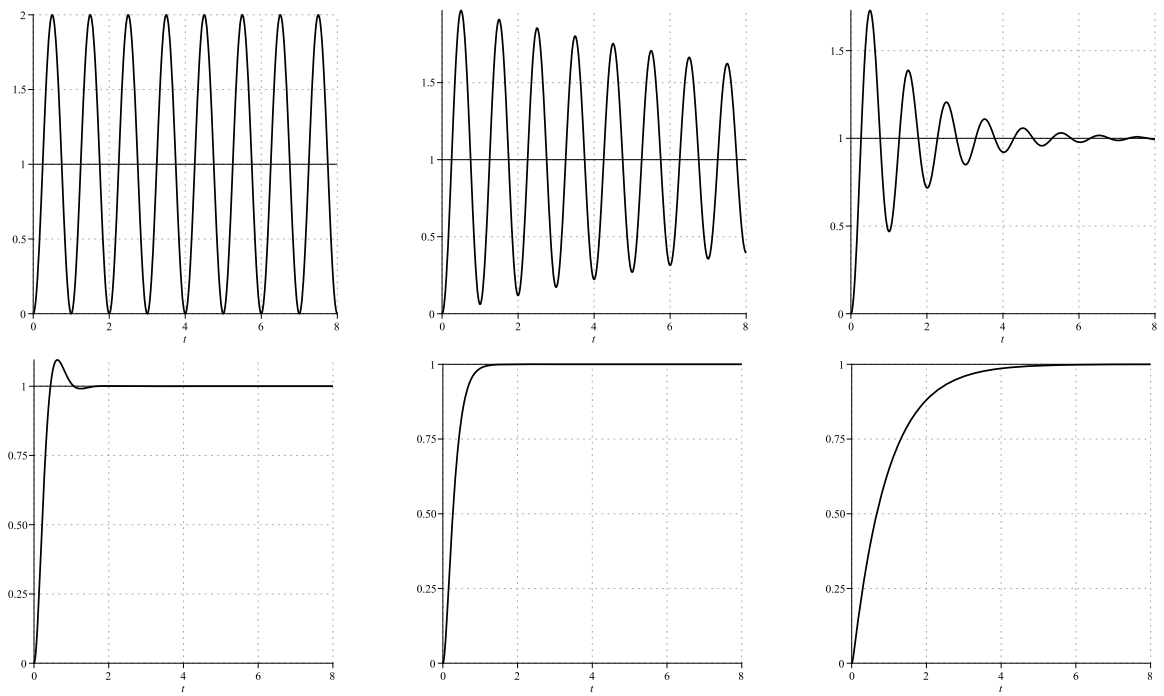


Figure 14.7: Response of the spring balance to a unit step-change input for $\xi = 0, 0.01, 0.1, 0.6, 1$ and 3 . The natural frequency is 1 Hz.